AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

APRIL/ MAY 2017

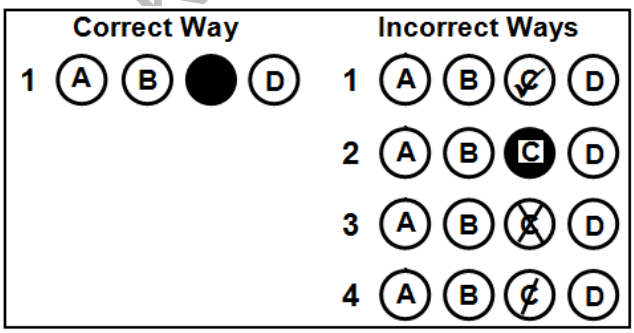
Mathematics Paper I

Time: 40 minutes Marks: 30

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. The highest common factor (H.C.F.) of $3(x+5)^2$ and $15(x+5)$ is

   A. $15(x+5)^2$
   B. $3(x+5)^2$
   C. $15(x+5)$
   D. $3(x+5)$

2. On simplification, $\frac{ab+b^2}{(a+b)^2}$ is equal to

   A. b
   B. $\frac{b}{a}$
   C. $\frac{b}{(a+b)}$
   D. $\frac{a}{(a+b)}$

3. The square root of $x^2 - \frac{2x}{a} + \frac{1}{a^2}$ is

   A. $x - \frac{2}{a}$
   B. $x - \frac{1}{a}$
   C. $x - \frac{1}{a^2}$
   D. $x - \sqrt{\frac{2x}{a}} + \frac{1}{a}$

4. Which of the following forms is suitable for the partial fraction of $\frac{5x+3}{(x-2)(x+3)}$?

   A. $\frac{A}{(x-2)} + \frac{B}{(x+3)}$
   B. $\frac{A+B}{(x-2)(x+3)}$
   C. $\frac{A}{(x-2)} + \frac{Bx+C}{(x+3)}$
   D. $\frac{Ax+B}{(x-2)} + \frac{C}{(x+3)}$

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5. The solution set of $\frac{x}{3} - b = 0$ is

A. $\{0\}$
B. $\{b\}$
C. $\{3b\}$
D. $\{b-3\}$

6. The solution set of $\frac{\sqrt{x+2}}{2} = 0$ is equal to

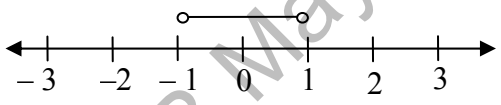
A. $\{1\}$
B. $\{2\}$
C. $\{-2\}$
D. $\{-2, 2\}$

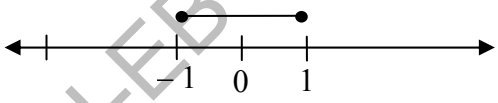
7. Two times of a number x reduced by 3 is 5. The equation representing the given situation is

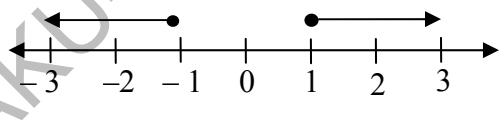
A. $x^2 - 3 = 5$
B. $2x - 3 = 5$
C. $2x + 3 = 5$
D. $2(x-3) = 5$

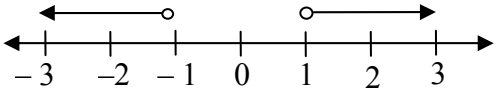
8. The solution set of $|x| < 1$ may be represented on the number line as follows:

(**Note:** $x \in R$)

A.


B.


C.


D.


9. For the equation $2x + 3y = 6,$ the missing value in the given table is

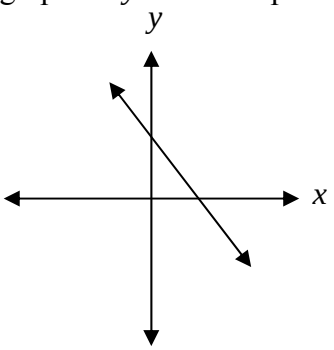
x	0	?
y	2	4

A. -9
B. -3
C. 3
D. 9

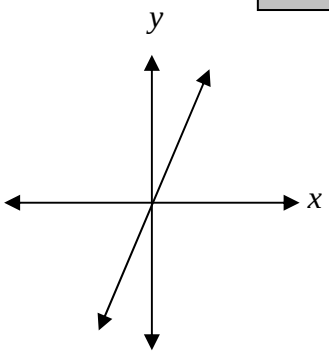
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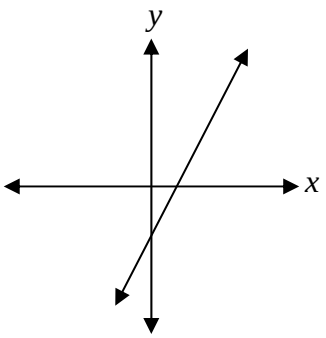
10. The graph of $y = 3x$ is represented by



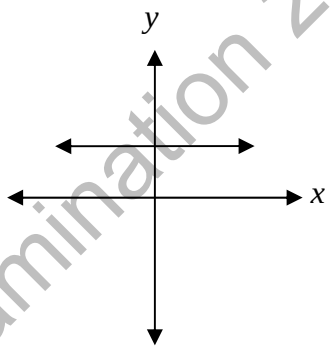
A



B



C



D

11. If the given ordered pairs $(a-2, b-3)$ and $(-2, -3)$ are equal, then

A. $a = 0$ and $b = 0$
B. $a = 2$ and $b = 3$
C. $a = 4$ and $b = 6$
D. $a = -2$ and $b = -3$

12. The standard form of the quadratic equation $\frac{3}{2} + x^2 = 2x$ is

A. $x^2 + 4x + 3 = 0$
B. $x^2 - 2x + 3 = 0$
C. $2x^2 + 4x + 3 = 0$
D. $2x^2 - 4x + 3 = 0$

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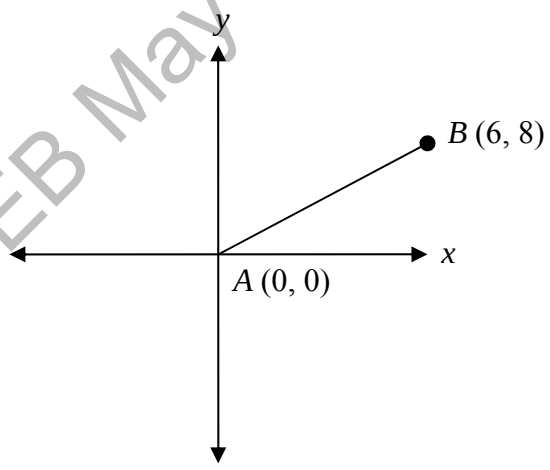
13. The solution set of the quadratic equation $x^2 = -\frac{1}{2}$, where $x \in R$, is

A. $\{\ \}$
B. $\{0\}$
C. $\left\{\pm\frac{1}{4}\right\}$
D. $\left\{\pm\frac{1}{\sqrt{2}}\right\}$

14. What should be added to $x^2 + 3x$ to make it a perfect square?

A. $\frac{1}{9}$
B. $\frac{3}{2}$
C. $\frac{9}{4}$
D. 9

15. In the given diagram, the length of line segment $\overline{AB}$ is



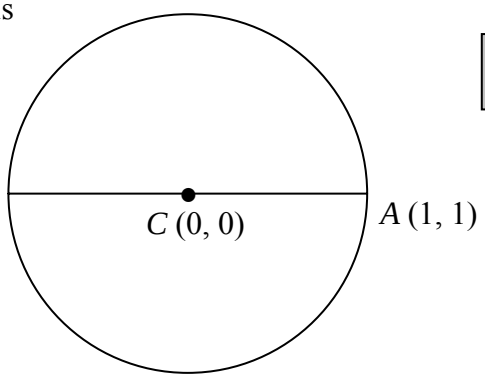
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A. 6
B. 7
C. 8
D. 10

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16. In the given diagram, C (0, 0) is the centre and A (1, 1) is a point on the circumference of the circle. The radius of the circle is

A. 2
B. 4
C. $\sqrt{2}$
D. $2\sqrt{2}$



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17. The distance between the points $(-3, 0)$ and $(0, -1)$ is

A. 8
B. $\sqrt{8}$
C. 10
D. $\sqrt{10}$

18. The angle $(53.25)^{\circ}$ in degree, minute and seconds is equal to

A. $53^{\circ}\ 15'0''$
B. $53^{\circ}\ 25'0''$
C. $53^{\circ}\ 20'5''$
D. $53^{\circ}\ 10'15''$

19. In the fourth quadrant, which of the following trigonometric ratios is positive?

A. $\sin\theta$
B. $\cos\theta$
C. $\tan\theta$
D. $\cot\theta$

20. On simplification of $\frac{1+\tan^2\theta}{\cos^2\theta}$, we get

A. 1
B. $\cos^2\theta$
C. $\sec^2\theta$
D. $\sec^4\theta$

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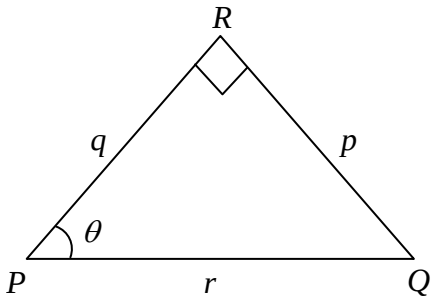
21. In the right-angled triangle PQR, $\sin^2\theta$ is equal to

A. $\frac{p}{r}$

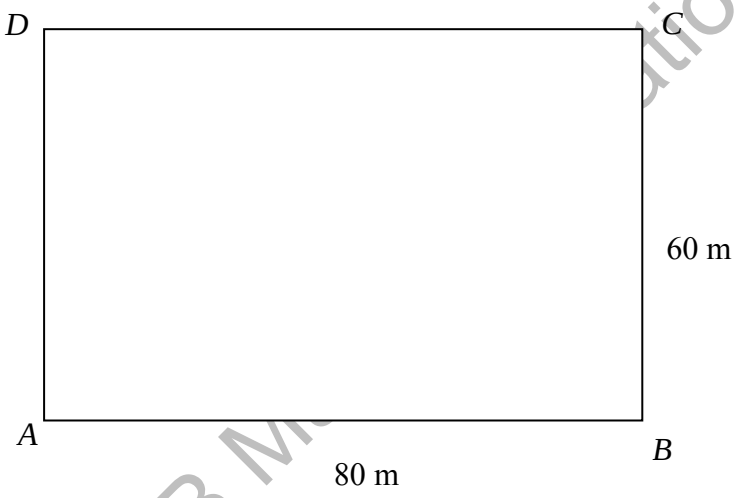
B. $\frac{p^2}{r^2}$

C. $\frac{p}{q}$

D. $\frac{p^2}{q^2}$



22. Sara and Sajid visited a park which is rectangular in shape as given in the following diagram. A tuck shop is located at point A and a coffee shop is located at point C. The shortest distance between tuck shop and coffee shop is



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A. $20\sqrt{7}$ m

B. 100 m

C. $2\sqrt{35}$ m

D. 140 m

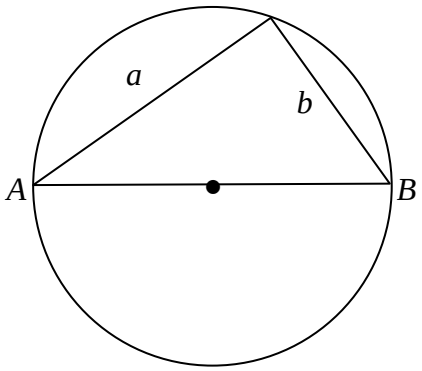
23. In the given diagram, if the diameter of the circle is AB, then radius of the circle in terms of a and b will be

A. $\frac{a^2-b^2}{2}$

B. $\frac{\sqrt{a^2-b^2}}{2}$

C. $\frac{a^2+b^2}{2}$

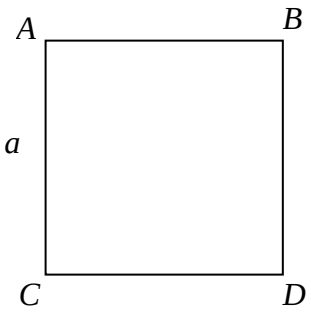
D. $\frac{\sqrt{a^2+b^2}}{2}$



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24. Consider the given square *ABCD*, the length of its diagonal *BC* is equal to

A. a^2
B. $2a$
C. $\sqrt{2}a$
D. $\sqrt{2}a^2$



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25. A circle has two chords of 5 cm and 6 cm that lie at a distance of 7 cm and *x* cm respectively from the centre of the circle. The value of *x* will be

A. less than 7 cm
B. equal to 7 cm
C. greater than 7 cm
D. between 7 cm and 11 cm

26. If the distance between centres of two circles touching each other externally is 10 cm and radius of one the circle is 7 cm then the radius of the other circle will be

A. 3 cm
B. 5 cm
C. 7 cm
D. 17 cm

27. If two circles touch each other externally and their radii are 8 cm and 12 cm, then the distance between their centres is

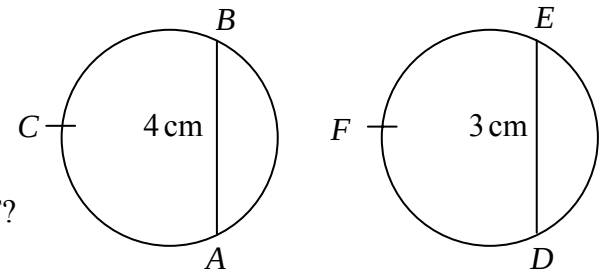
A. 2 cm
B. 4 cm
C. 10 cm
D. 20 cm

28. For the given two congruent circles, consider the following statements.

I. Arc *AB* < Arc *DE*
II. Arc *ACB* > Arc *DFE*
III. Arc *ACB* < Arc *DFE*

Which of the following options is CORRECT?

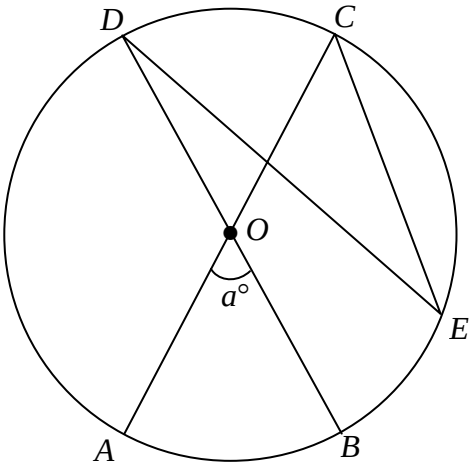
A. II only
B. III only
C. I and II
D. I and III



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29. In the given circle with centre O which of the following options is CORRECT?

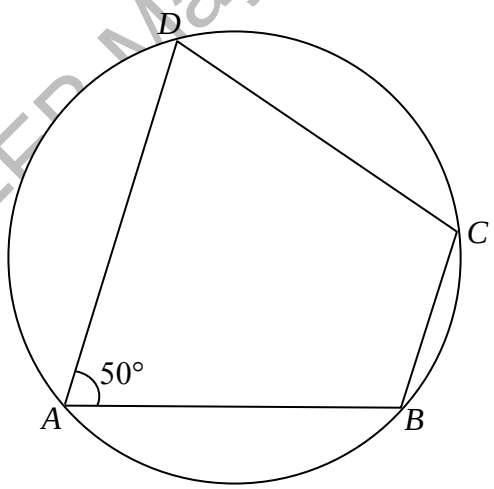


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I. $m\angle AOB = m\angle DOC$
II. $m\angle AOB = 2m\angle DEC$
III. $m\angle DOC = \frac{1}{2}m\angle DEC$

A. I only
B. II only
C. I and II
D. I and III

30. If $ABCD$ is a quadrilateral inscribed in a circle and $m\angle A = 50^o$, then $m\angle C$ is equal to



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A. 40°
B. 50°
C. 90°
D. 130°

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